

William Kojumian

AI & Software Engineer | LLM Systems & Full-Stack | Robotics

☎ +374 44 231 297 — ✉ WilliamKoju@gmail.com — 🔗 [linkedin.com/in/williamkojumian](https://www.linkedin.com/in/williamkojumian)
🐙 github.com/twillur — 🌐 twillur.github.io — 📍 Yerevan, Armenia

Professional Summary

AI and software engineer who ships production LLM systems end to end — RAG, multi-agent orchestration, natural-language-to-SQL, and workflow automation — on a full .NET/C# and Python stack. Delivered a multi-tenant enterprise AI product used by live clients at a Dubai AI firm, and independently built and pitched a multi-agent automation platform to the executives of MENA's largest professional AV distributor. American University of Armenia graduate (four years in Yerevan) returning to the Armenian tech ecosystem; MSc Robotics (Distinction track) researching LLM-grounded task planning for autonomous robot navigation (ROS 2, Nav2, Gazebo).

Professional Experience

AI Engineering Intern

Mar 2026 – Jul 2026

Enhance Intelligence & Technologies | Dubai, UAE

- Built and shipped **IntelliChat**, a production multi-tenant AI analytics assistant (Blazor Server, .NET/C#, Oracle 23ai) that translates natural language into SQL across 9 business domains, serving live enterprise ERP clients with ~80% end-to-end query success on real data.
- Designed a two-layer semantic cache — Redis exact-match L0 over **Oracle 23ai vector similarity search** — and embedding-based domain routing that cut prompt size by 50–60% and let repeated questions skip LLM generation entirely, reducing per-query cost and latency.
- Implemented enterprise-grade security: Oracle VPD row-level tenant isolation, Google OAuth 2.0 + JWT with silent refresh, role-based access control, AES-256 credential encryption, and signed client-side session state (fixed a live privilege-escalation vector).
- Built an agentic self-improvement loop: nightly **Hangfire** jobs mine chat history to retrain the query cache and auto-generate per-tenant business insights, with a human-in-the-loop UI for correcting failed queries — zero manual intervention in steady state.
- Added bring-your-own-key multi-LLM support (OpenAI GPT-4o, Groq, DeepSeek), per-tenant token metering, self-serve onboarding with email verification, and PWA packaging; iterated directly with the client through weekly stakeholder reviews.

AI Solutions Consultant (Independent)

May 2026 – Jun 2026

Advanced Media Trading LLC | Dubai, UAE

- Independently designed and built a **multi-agent workflow-automation platform** for MENA's largest professional AV distributor (~94 employees, DJI authorized): **LangGraph** state-machine routing across four department agents (Sales, Finance, Distribution, Service), **LangChain** structured tools, and GPT-4o function calling over an SAP-aligned ERP dataset (20+ tables).
- Implemented semantic product search with **LlamaIndex** + **ChromaDB** (RAG over a 42-product vector store with t-SNE embedding visualization), pandas-driven financial reporting (aging, receivables, VAT by country), and live **n8n** automations firing real emails for service tickets and RFQ-to-quote generation.
- Presented the working product to company sector heads as a proposed first-AI-hire initiative; the demo led to ongoing hiring discussions for an internal AI automation role.

Robotics & AI Engineer

2025

InspireTech | Dubai, UAE

- Supported motion-control development and stability testing for **Unitree humanoid and quadruped** robot platforms; provided hands-on troubleshooting and real-time calibration in a fast-paced R&D environment.
- Operated and demonstrated advanced robotic systems live at Doer's Summit 2025, showcasing human-robot interaction, movement sequencing, and multi-robot coordination to public and industry audiences.

Technical Lead & Club Organizer

2021 – 2025

Drone Club, American University of Armenia | Yerevan, Armenia

- Founded workshops and led the university UAV community for 4 years: flight physics, drone assembly, simulation workflows, and hands-on flight training; directed student teams through practical engineering and system-testing missions.

Education

MSc Robotics Engineering

Sep 2025 – Oct 2026 (expected)

Middlesex University Dubai (Middlesex University London), UAE

Expected: Distinction

Dissertation: Natural Language Task Planning for Autonomous Ground Robot Navigation Using Large Language Models

Core Modules: AI & Machine Learning, Advanced Control Systems, Robot Programming, Automation

BSc Engineering Science (Mechatronics)

Sep 2021 – Jun 2025

American University of Armenia, Yerevan

Key Areas: Circuits, Embedded Systems, Electromechanics, Control Theory, CAD, Programming

Key Projects

LLM-to-Nav2 Task Planner (MSc Dissertation)

2026 – Present

- Building a prompt-engineered pipeline that grounds natural-language commands into structured, schema-validated **Nav2 waypoint plans** for an autonomous warehouse patrol robot (ROS 2 Humble + Gazebo); provider-agnostic planner already smoke-tested end to end.
- Designed a 100-command evaluation benchmark across 5 complexity levels — including deliberate safety traps the planner must refuse — with an automated evaluation harness; documented as a public research blog ([project blog](#)).

Visual Quality Inspection — CNN Defect Classifier

2025

- Fine-tuned a **MobileNetV2** transfer-learning model on a custom image dataset for automated PASS/FAIL packaging inspection, achieving ~95% validation accuracy across the full pipeline (collection, augmentation, training, evaluation); [source on GitHub](#).

Sandsifter — Autonomous Beach-Cleaning Robot

2025 – 2026

- Developed a full ROS 2 (Humble) + Gazebo simulation of an autonomous beach-cleaning robot with a `ros2_control` manipulator arm for debris collection.
- Implemented a custom boustrophedon (lawnmower-pattern) coverage path planner for complete-area traversal with sensor-driven obstacle avoidance.

Earlier Projects

2023 – 2025

- **Epson RC+ industrial automation:** programmed an Epson arm for autonomous pick-and-place with sensor-driven sorting and trajectory optimization (2025). **BSc capstone:** designed and prototyped a UAV intralogistics system for autonomous payload transport (2023–24).

Technical Skills

- **LLM & Agentic AI:** GPT-4o / OpenAI API, prompt engineering, RAG, multi-agent orchestration (LangGraph), LangChain, LlamaIndex, function calling / tool use, vector databases (ChromaDB, Oracle 23ai, Redis), embeddings & semantic search, natural-language-to-SQL, LLM evaluation
- **Machine Learning:** PyTorch, transfer learning (MobileNetV2), computer vision, XGBoost, supervised/unsupervised learning, model evaluation
- **Programming & Tools:** Python, C++, C# (.NET / Blazor Server), SQL, MATLAB, Git/GitHub, Linux/Ubuntu, Docker (basic), Flask, REST APIs, n8n, Hangfire, OAuth 2.0/JWT, $\text{\LaTeX}$
- **Robotics:** ROS 2 (Humble), Nav2, Gazebo, `ros2_control`, motion control, sensor integration & perception, embedded systems, UAV systems, Unitree platforms, Epson RC+, CAD (Fusion 360)
- **Data:** pandas, NumPy, Oracle SQL, SQLite, feature engineering, data visualization
- **Languages:** English (native fluency), Armenian (professional), Arabic (professional)

References

Satenik Mnatsakanyan — Program Chair, Engineering Science, American University of Armenia — smnatsakanyan@aua.am

Sameer Kishore — Campus Program Coordinator (Robotics), Middlesex University Dubai — s.kishore@mdx.ac.ae